The Exam Hall Seat Duplication Checker

Week 1 Assignment - Problem 1 | Java

Problem Summary

Scan an integer array using nested loops only (no Collections classes). Print each duplicated seat number once, or confirm that no duplicates exist.

Java Solution

```
public class Main {
    static void checkDuplicateSeats(int[] seatNumbers) {
        boolean found = false;
        for (int i = 0; i < seatNumbers.length; i++) {
            boolean alreadyReported = false;
            for (int k = 0; k < i; k++) {
                if (seatNumbers[k] == seatNumbers[i]) {
                    alreadyReported = true;
                    break;
                }
            }
            if (alreadyReported) continue;
            for (int j = i + 1; j < seatNumbers.length; j++) {
                if (seatNumbers[i] == seatNumbers[j]) {
                    System.out.println("Duplicate Seat Number Found: " + seatNumbers[i]);
                    found = true;
                    break;
                }
            }
        }
        if (!found) System.out.println("No Duplicate Seats Found");
    }
    public static void main(String[] args) {
        checkDuplicateSeats(new int[]{101, 102, 103, 102, 105});
        checkDuplicateSeats(new int[]{101, 102, 103, 104, 105});
    }
}
```

Sample Output

```
Duplicate Seat Number Found: 102
No Duplicate Seats Found
```

Complexity / Key Notes

Time: O(n^2). Extra space: O(1). No Collections classes are used. Duplicate values are printed only once.